

HDOC DAY-27 ASSIGNMENT

Menu-Driven Pyramid/Diamond Patterns Using Switch-Case

1. Problem Understanding

The task is to write a menu-driven C program using switch-case to print five pyramid/diamond patterns. Nested loops are used without arrays. Each menu option is implemented in a separate parameterless function with void return type. Each function reads the number of rows and prints the selected pattern.

- Full pyramid
- Inverted pyramid
- Diamond pattern
- Palindromic number pyramid
- Repeated-number pyramid

2. Standard Test Cases

Test Case	Choice	Input	Expected Output	Status
TC1	1	4	Full pyramid with 4 rows	Pass
TC2	2	4	Inverted pyramid with 4 rows	Pass
TC3	3	4	Diamond with 7 rows	Pass
TC4	4	4	Palindromic pyramid with 4 rows	Pass
TC5	5	4	Repeated-number pyramid with 4 rows	Pass

3. Algorithm

A. Common Menu Logic

1. Start.
2. Display the five pattern options.
3. Read the user's choice.
4. Use switch-case to select the required pattern.
5. Call the corresponding void function.
6. The selected function reads n and prints the pattern using nested loops.
7. If the choice is invalid, display "Invalid choice".
8. Stop.

B. Full Pyramid

1. Start.
2. Read the number of rows n.
3. For each row i from 1 to n, print n - i leading spaces.
4. Print $2i - 1$ asterisks separated by spaces.
5. Move to the next line after each row.
6. Stop.

C. Inverted Pyramid

1. Start.
2. Read the number of rows n .
3. For each row i from n down to 1, print $n - i$ leading spaces.
4. Print $2i - 1$ asterisks separated by spaces.
5. Move to the next line after each row.
6. Stop.

D. Diamond Pattern

1. Start.
2. Read n for the upper half.
3. Print the upper half from row 1 to n using $n - i$ spaces and $2i - 1$ asterisks.
4. Print the lower half from row $n - 1$ down to 1 using $n - i$ spaces and $2i - 1$ asterisks.
5. Move to a new line after every row.
6. Stop.

E. Palindromic Number Pyramid

1. Start.
2. Read the number of rows n .
3. For each row i from 1 to n , print $n - i$ leading spaces.
4. Print numbers from 1 to i in increasing order.
5. Print numbers from $i - 1$ down to 1 in decreasing order.
6. Move to the next line after each row.
7. Stop.

F. Repeated-Number Pyramid

1. Start.
2. Read the number of rows n .
3. For each row i from 1 to n , print $n - i$ leading spaces.
4. Print the row number i exactly $2i - 1$ times.
5. Move to the next line after each row.
6. Stop.

4. C Program

```
#include <stdio.h>
void fullPyramid(){
    int n,i,j;
    printf("Enter number of rows: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++){
        for(j=1;j<=n-i;j++)
            printf(" ");
        for(j=1;j<=2*i-1;j++)
            printf("* ");
        printf("\n");
    }
}
```

```

void invertedPyramid(){
    int n,i,j;
    printf("Enter number of rows: ");
    scanf("%d",&n);
    for(i=n;i>=1;i--){
        for(j=1;j<=n-i;j++)
            printf(" ");
        for(j=1;j<=2*i-1;j++)
            printf("* ");
        printf("\n");
    }
}

void diamond(){
    int n,i,j;
    printf("Enter number of rows: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++){
        for(j=1;j<=n-i;j++)
            printf(" ");
        for(j=1;j<=2*i-1;j++)
            printf("* ");
        printf("\n");
    }
    for(i=n-1;i>=1;i--){
        for(j=1;j<=n-i;j++)
            printf(" ");
        for(j=1;j<=2*i-1;j++)
            printf("* ");
        printf("\n");
    }
}

void palindromicPyramid(){
    int n,i,j;
    printf("Enter number of rows: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++){
        for(j=1;j<=n-i;j++)
            printf(" ");
        for(j=1;j<=i;j++)
            printf("%d ",j);
        for(j=i-1;j>=1;j--)
            printf("%d ",j);
        printf("\n");
    }
}

void repeatedNumberPyramid(){
    int n,i,j;

```

```

printf("Enter number of rows: ");
scanf("%d",&n);
for(i=1;i<=n;i++){
    for(j=1;j<=n-i;j++)
        printf(" ");
    for(j=1;j<=2*i-1;j++)
        printf("%d ",i);
    printf("\n");
}
}
int main(){
    int choice;
    printf("MENU\n");
    printf("1. Full pyramid\n");
    printf("2. Inverted pyramid\n");
    printf("3. Diamond pattern\n");
    printf("4. Palindromic number pyramid\n");
    printf("5. Repeated-number pyramid\n");
    printf("Enter your choice: ");
    scanf("%d",&choice);
    switch(choice){
        case 1:
            fullPyramid();
            break;
        case 2:
            invertedPyramid();
            break;
        case 3:
            diamond();
            break;
        case 4:
            palindromicPyramid();
            break;
        case 5:
            repeatedNumberPyramid();
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}

```

5. Compilation and Execution Evidence

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)-[~/Desktop]
$ gedit question.c

** (gedit:2227): WARNING **: 15:36:51.112: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2227): WARNING **: 15:36:51.112: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)-[~/Desktop]
$ gcc question.c -o question

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 1
Enter number of rows: 4
  *
 * *
* * *
* * * *
* * * * *

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 2
Enter number of rows: 4
* * * * *
 * * *
  * *
   *
  
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 3
Enter number of rows: 4
  *
 * *
* * *
* * * *
* * * * *
 * *
  *

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 4
Enter number of rows: 4
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid

```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Enter number of rows: 4
*
* *
* * *
* * * *
* * * * *
* * * * *
* * *
* *
*

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 4
Enter number of rows: 4
1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1

(kali@kali)-[~/Desktop]
$ ./question
MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 5
Enter number of rows: 4
1
2 2 2
3 3 3 3 3
4 4 4 4 4 4 4

(kali@kali)-[~/Desktop]
$
```

6. Verification of All Test Cases

Test Case	Choice	Input	Expected Result	Actual Result	Status
TC1	1	4	Full pyramid	Full pyramid	Pass
TC2	2	4	Inverted pyramid	Inverted pyramid	Pass
TC3	3	4	Diamond pattern	Diamond pattern	Pass
TC4	4	4	Palindromic pyramid	Palindromic pyramid	Pass
TC5	5	4	Repeated-number pyramid	Repeated-number pyramid	Pass
